

AnalystLab Africa Consulting

Project Management Research

1. What Is a Project and Project Management?

A project is a temporary undertaking with a defined start and end, carried out to produce a unique product, service, or result. This distinguishes it from operations, which are ongoing and repetitive. Because a project is temporary and produces something new, it typically involves more uncertainty than routine operational work, which is part of why a dedicated management discipline exists for it.

Project management is the application of knowledge, skills, tools, and techniques to project activities in order to meet defined requirements. It involves balancing competing constraints, commonly described as the “triple constraint” of scope, time, and cost, with quality often added as a fourth dimension. Changing one constraint typically affects at least one of the others; for example, expanding scope without adjusting time or cost usually reduces quality or requires renegotiating the schedule and budget.

2. The Project Life Cycle and the Five Process Groups

The Project Management Institute's PMBOK Guide organises project work into five process groups: Initiating, Planning, Executing, Monitoring & Controlling, and Closing. A process group is a category of related project management activities, not a fixed calendar phase in particular, Monitoring & Controlling runs continuously across the life of the project rather than occupying a single block of time.

Process Group	Purpose	Typical Outputs
Initiating	Formally define and authorise a new project or phase; align stakeholders on purpose and high-level scope	Project Charter, initial stakeholder identification
Planning	Develop the roadmap covering scope, schedule, cost, quality, risk, and communication	WBS, schedule, budget, risk register, communication plan
Executing	Coordinate people and resources to carry out the plan and produce deliverables	Completed work packages, project deliverables
Monitoring & Controlling	Track performance against the plan, identify variances, and manage changes; runs continuously alongside the other groups	Status reports, updated risk log, change requests
Closing	Formally complete and hand over the project or phase, and capture lessons learned	Sign-off, handover documentation, lessons-learned log

The project life cycle refers to the series of phases a project passes through from start to finish (for example, a construction project might have design, procurement, construction, and commissioning phases). The five process groups are distinct from this: they can, and typically

do, repeat within each phase of a multi-phase project. It is also worth distinguishing process groups from a Work Breakdown Structure (WBS): a WBS is meant to be deliverable-oriented, decomposing the project into the tangible outputs it produces, rather than organised around which process group created them.

3. Project Management Methodologies

Beyond the PMBOK process groups, several named methodologies and frameworks guide how project management is actually carried out in practice. The three most commonly referenced are PMBOK/PMI practice, PRINCE2, and Agile.

Methodology	Origin / Best Fit	Key Characteristic
PMBOK (PMI)	General-purpose knowledge base used across industries; adaptable to hybrid or Agile use	A guide of knowledge areas and process groups rather than a rigid step-by-step method
PRINCE2	Originated in UK government; widely used for large-scale, regulated, or public-sector projects	Process-driven with clearly defined roles, stages, and formal sign-off gates (7 Principles, 7 Themes, 7 Processes)
Agile / Scrum	Software and product development; environments with evolving requirements	Iterative delivery in short cycles (sprints) rather than a single upfront plan

These are not always mutually exclusive. In practice, many organisations blend approaches for example, using PRINCE2-style governance for overall project sign-off while allowing Agile sprints within individual workstreams, or applying PMBOK knowledge areas (such as risk and procurement) around an otherwise iterative delivery process. The choice of methodology generally depends on how well-defined the requirements are upfront, how much governance and auditability the project needs, and the industry or sector norms involved.

4. Core Project Management Knowledge Areas

The PMBOK Guide also organises project management practice into ten knowledge areas, which cut across the five process groups. These represent the different dimensions of a project that need active management throughout its life cycle.

Knowledge Area	What It Covers
Scope Management	Defining and controlling what is, and is not, included in the project
Schedule Management	Sequencing activities and managing the timeline
Cost Management	Estimating, budgeting, and controlling project costs
Quality Management	Ensuring deliverables meet the required standards
Resource Management	Acquiring and managing the people and physical resources needed
Communications Management	Ensuring the right information reaches the right people at the right time

Knowledge Area	What It Covers
Risk Management	Identifying, analysing, and responding to project risks
Procurement Management	Acquiring goods and services from outside the project team
Stakeholder Management	Identifying stakeholders and managing their expectations and engagement
Integration Management	Coordinating all other knowledge areas into a coherent whole

5. Common Project Management Deliverables

Certain artefacts recur across most project management approaches, regardless of methodology:

- Project Charter - formally authorises the project and names the project manager
- Stakeholder Register - identifies who is affected by or can affect the project, and how to engage them
- Work Breakdown Structure (WBS) - decomposes the project into manageable, deliverable-oriented components
- Schedule / Gantt Chart - sequences tasks against time
- Risk Register - logs identified risks, their likelihood and impact, and planned responses
- Communication Plan - defines who needs what information, how often, and through which channel

References

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